# Random Variables

**Definition**: A **random variable (abbreviated r.v.)**, and denoted $X$, is a numerical quantity whose value is determined by a random experiment.

**Examples:**

- Two dice are rolled, and $X$ is the sum of the two upward faces.
- We count the number of earthquakes, $X$, that occur in the San Francisco region from 2000 to 2050.
- Today the DJI index opened at 51,415.55, and $X$ is the value of the index in thirty days.

# Example 1

A user is shown a sequence of 3 targeted ads during a web session. The probability of the user clicking the first ad is 5%. However, if they click an ad, the probability of them clicking the *next* ad increases to 15% (and drops back to 5% if they ignore an ad).

$X$= the total number of ads clicked during the session

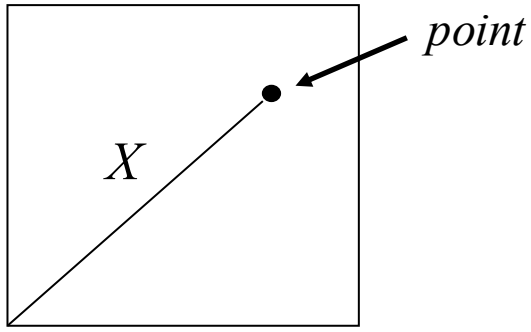
| $s$ | $X$ |
|---|---|
| NNN | 0 |
| CNN | 1 |
| NCN | 1 |
| NNC | 1 |
| NCC | 2 |
| CCN | 2 |
| CNC | 2 |
| CCC | 3 |

Q: How likely is each unique value of $X$?

# Example 2

A point is selected at random from the unit square (whose sides are of length 1).

Let $X$ be the **distance** from the point to the lower left corner



# Exercise 1

An automated hyperparameter tuning script evaluates a machine learning model's validation loss after each training epoch. Compared to the previous epoch, the loss can result in three discrete outcomes: **Decrease (D)**, **Increase (I)**, or remain **Unchanged (U)**.

To prevent overfitting and save compute resources, the script uses a multi-condition early stopping callback. The training terminates immediately if either of the following occurs:

The loss **Increases** on two *consecutive* epochs.

The loss remains **Unchanged** on three *consecutive* epochs.

Let $N$ denote the total number of epochs evaluated until the training process terminates. State the three smallest possible values of $N$ and list all exact sequences of epoch outcomes corresponding to each of these values.

## Definition –probability mass function, $p(x)$, of a discrete random variable $X$.

For a discrete random variable, $X$, and any real number, $x$, we define the probability mass function (pmf) as

$$p(x) = P(X = x)$$

Note:

- The random variable (as function defined on the sample space) is denoted $X$.
- The values it can take (real numbers) are denoted as $x \in \mathbb{R}$

# Examples

Two dice are rolled, and $X$ is the sum of the two upward faces. $S$, sample space is shown below with the value of $X$ for each outcome

| | | | | | |
|---|---|---|---|---|---|
| (1,1)<br>2 | (1,2)<br>3 | (1,3)<br>4 | (1,4)<br>5 | (1,5)<br>6 | (1,6)<br>7 |
| (2,1)<br>3 | (2,2)<br>4 | (2,3)<br>5 | (2,4)<br>6 | (2,5)<br>7 | (2,6)<br>8 |
| (3,1)<br>4 | (3,2)<br>5 | (3,3)<br>6 | (3,4)<br>7 | (3,5)<br>8 | (3,6)<br>9 |
| (4,1)<br>5 | (4,2)<br>6 | (4,3)<br>7 | (4,4)<br>8 | (4,5)<br>9 | (4,6)<br>10 |
| (5,1)<br>6 | (5,2)<br>7 | (5,3)<br>8 | (5,4)<br>9 | (5,5)<br>10 | (5,6)<br>11 |
| (6,1)<br>7 | (6,2)<br>8 | (6,3)<br>9 | (6,4)<br>10 | (6,5)<br>11 | (6,6)<br>12 |

$$p(2) = P[X = 2] = P\left[\{(1,1)\}\right] = \frac{1}{36}$$

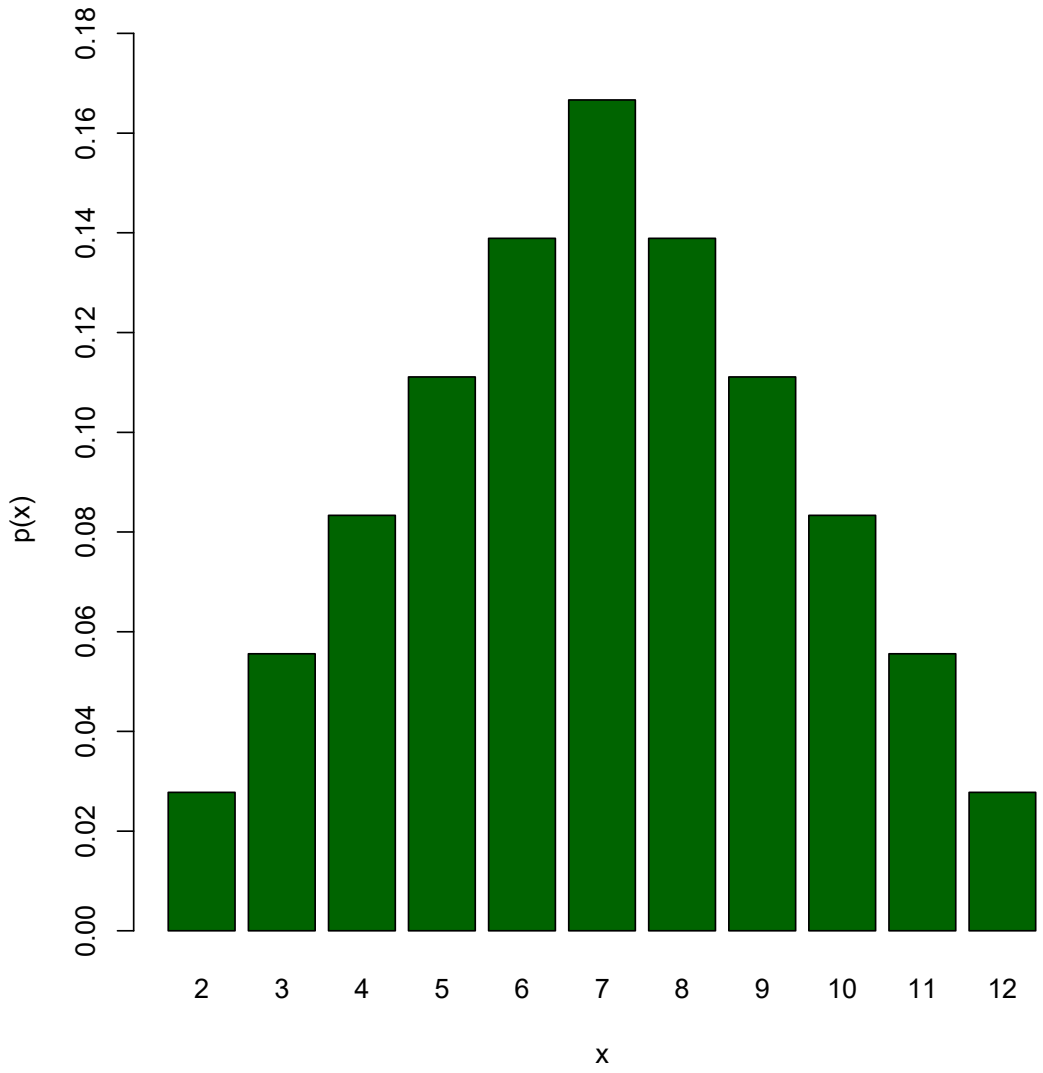
$$p(3) = P[X = 3] = P\left[\{(1,2),(2,1)\}\right] = \frac{2}{36}$$

$$p(4) = P[X = 4] = P\left[\{(1,3),(2,2),(3,1)\}\right] = \frac{3}{36}$$

$$p(5) = \frac{4}{36}, p(6) = \frac{5}{36}, p(7) = \frac{6}{36}, p(8) = \frac{5}{36}, p(9) = \frac{4}{36}$$

$$p(10) = \frac{3}{36}, p(11) = \frac{2}{36}, p(12) = \frac{1}{36}$$

and $p(x) = 0$ for all other $x$


p(x)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
0.18
2
3
4
5
6
7
8
9
10
11
12
x

# Cumulative Distribution Function (CDF)

The **CDF** of a random variable $X$, denoted as $F_X(x)$, is the function that specifies the probability that $X$ takes on a value less than or equal to $x$.

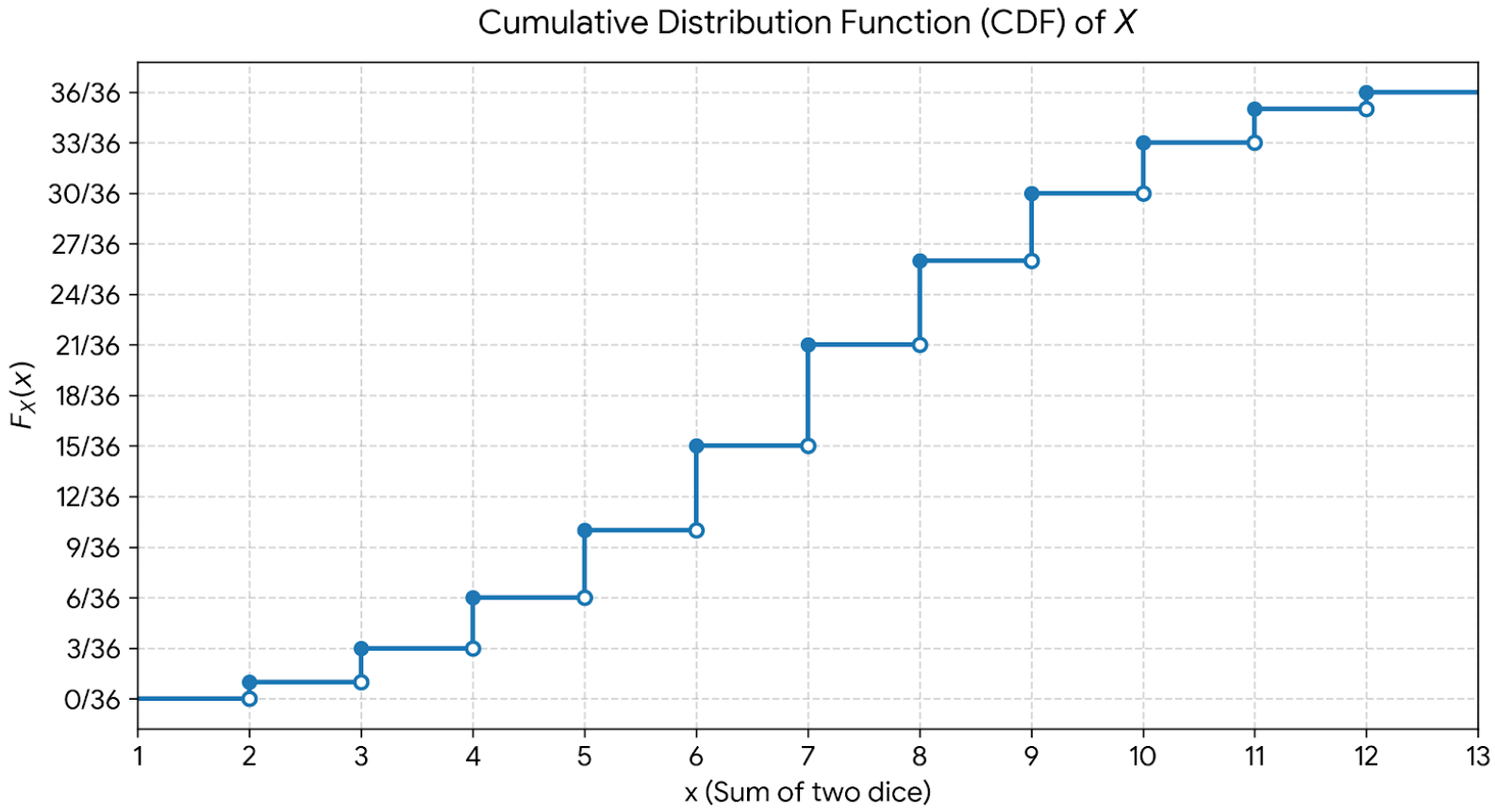
$$F_X(x) = P(X \leq x)$$

Unlike the Probability Density Function (PDF), which gives the relative likelihood of a continuous random variable taking on a specific point value, the CDF evaluates accumulated probability up to a threshold $x$.

**Core Mathematical Properties**

- **Probability Bounds:** $0 \leq F_X(x) \leq 1$for all $x \in \mathbb{R}$.
- **Asymptotic Limits:** $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$
- **Monotonic Non-Decreasing:** If $a \leq b$, then $F_X(a) \leq F_X(b)$.
- **Right-Continuity:** $\lim_{t \to x^+} F_X(t) = F_X(x)$.

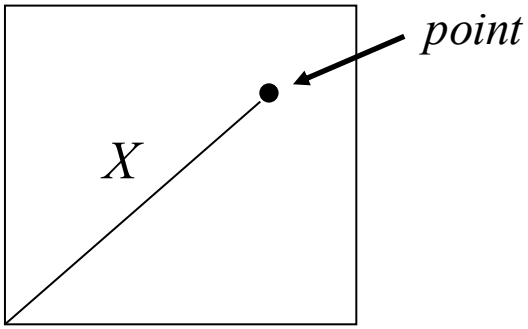
# Example: Rolling two dice




*Note:* $F(x)$ is a step function

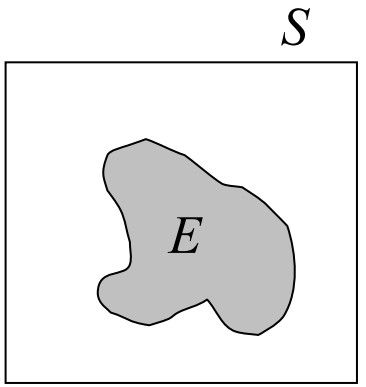
# Examples of random variables - continued

A point is selected at random from a square whose sides are of length 1. $X$ is the **distance** of the point from the lower left-hand corner.



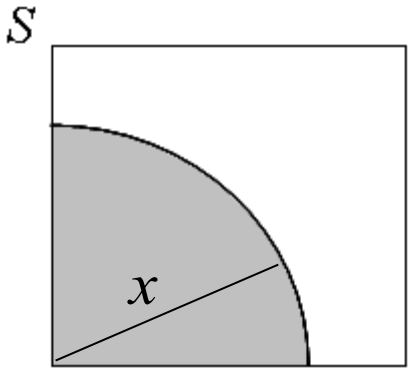

An event, $E$, is any subset of the square, $S$.

$P[E]$ = (area of $E$)/(Area of $S$) = area of $E$

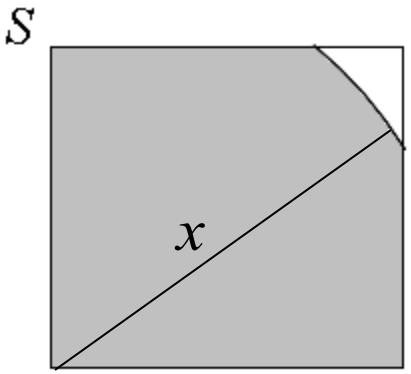


## The Cumulative distribution function

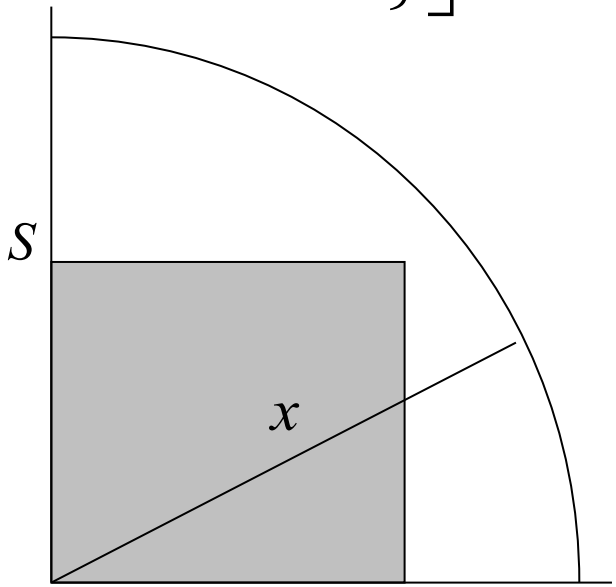
$$F(x) = P[X \le x] = P\left[\left\{\begin{array}{l}\text{set of all points within a} \\ \text{dist } x \text{ from lower left corner}\end{array}\right\}\right]$$




$0 \le x \le 1$

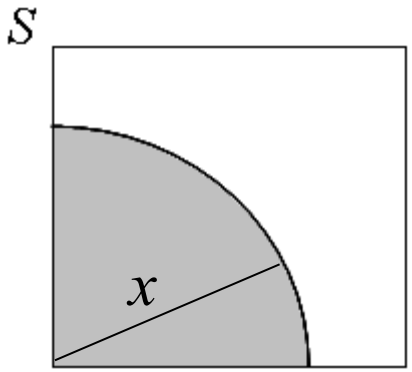



$1 \le x \le \sqrt{2}$

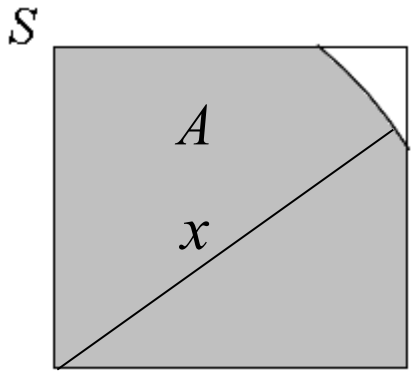



$\sqrt{2} \le x$

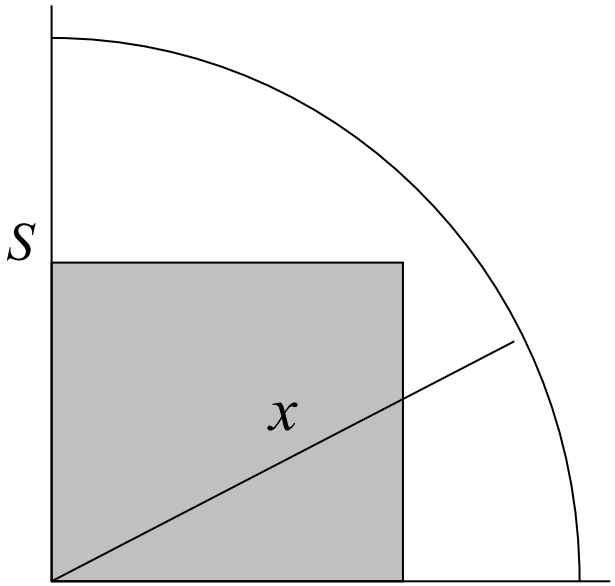
$$F(x) = P[X \le x] = \begin{cases} 0 & x \le 0 \\ \frac{\pi x^2}{4} & 0 \le x \le 1 \\ \text{Area } A & 1 \le x \le \sqrt{2} \\ 1 & \sqrt{2} \le x \end{cases}$$




$0 \le x \le 1$

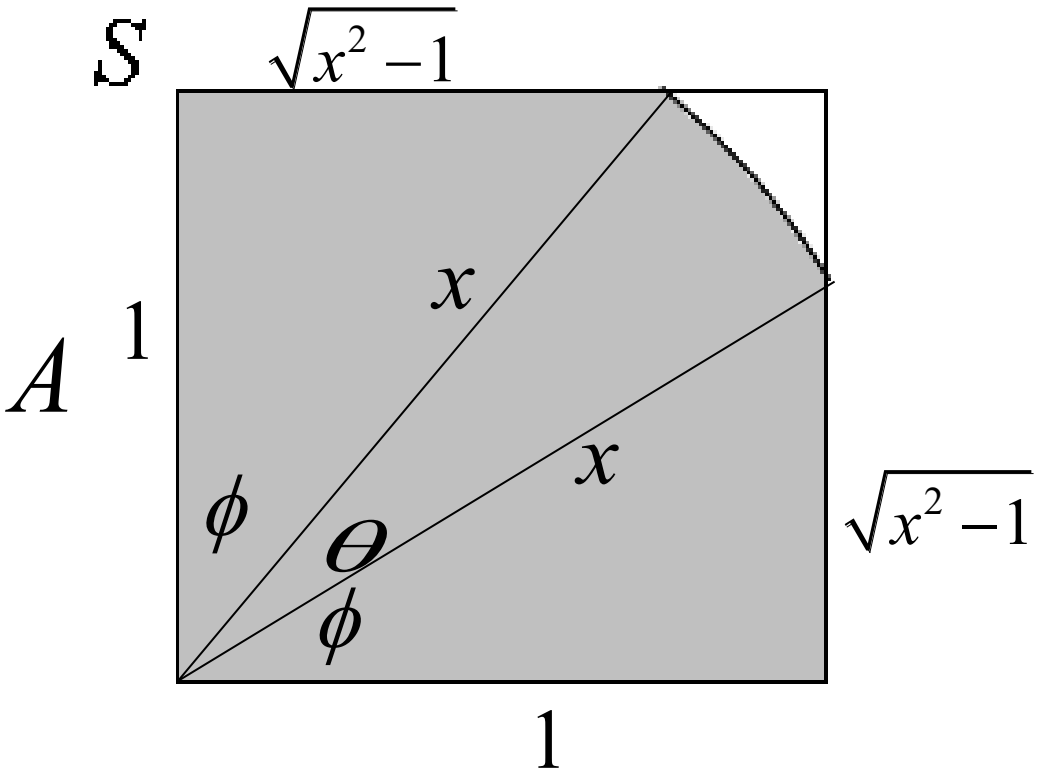



$1 \le x \le \sqrt{2}$




$\sqrt{2} \le x$

# Computation of Area $A$ $\quad 1 \le x \le \sqrt{2}$




$$\theta = \frac{\pi}{2} - 2\phi$$

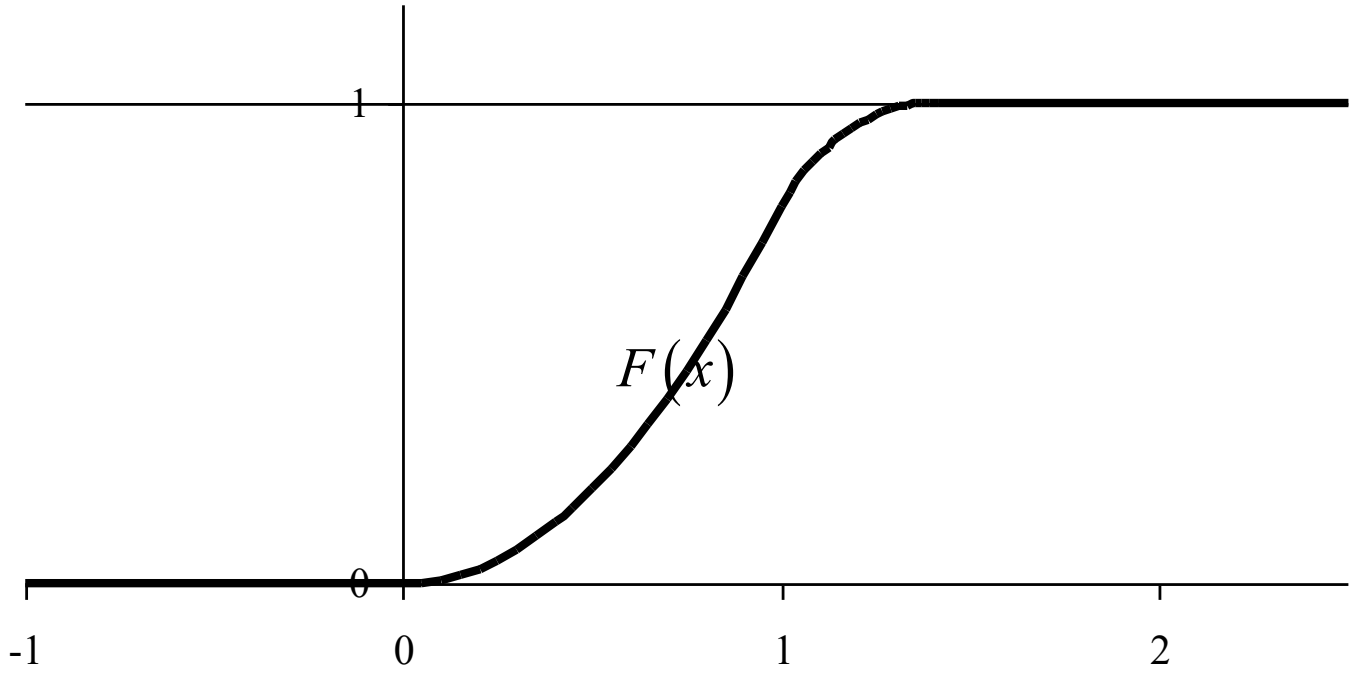
$$\tan(\phi) = \sqrt{x^2-1}$$

$$\phi = \tan^{-1}\left(\sqrt{x^2-1}\right)$$

$$A = 2\left(\frac{1\sqrt{x^2-1}}{2}\right) + \frac{\theta}{2\pi}\left(\pi x^2\right) = \sqrt{x^2-1} + \frac{\theta}{2}x^2$$

$$= \sqrt{x^2-1} + \left(\frac{\pi}{4} - \phi\right)x^2 = \sqrt{x^2-1} + \left(\frac{\pi}{4} - \tan^{-1}\left(\sqrt{x^2-1}\right)\right)x^2$$

$$F(x) = P[X \le x] = \begin{cases} 0 & x \le 0 \\ \dfrac{\pi x^2}{4} & 0 \le x \le 1 \\ \sqrt{x^2-1} + \left( \dfrac{\pi}{4} - \tan^{-1}\left(\sqrt{x^2-1}\right) \right) x^2 & 1 \le x \le \sqrt{2} \\ 1 & \sqrt{2} \le x \end{cases}$$



# Probability Density Function

Suppose that $X$ is a random variable.

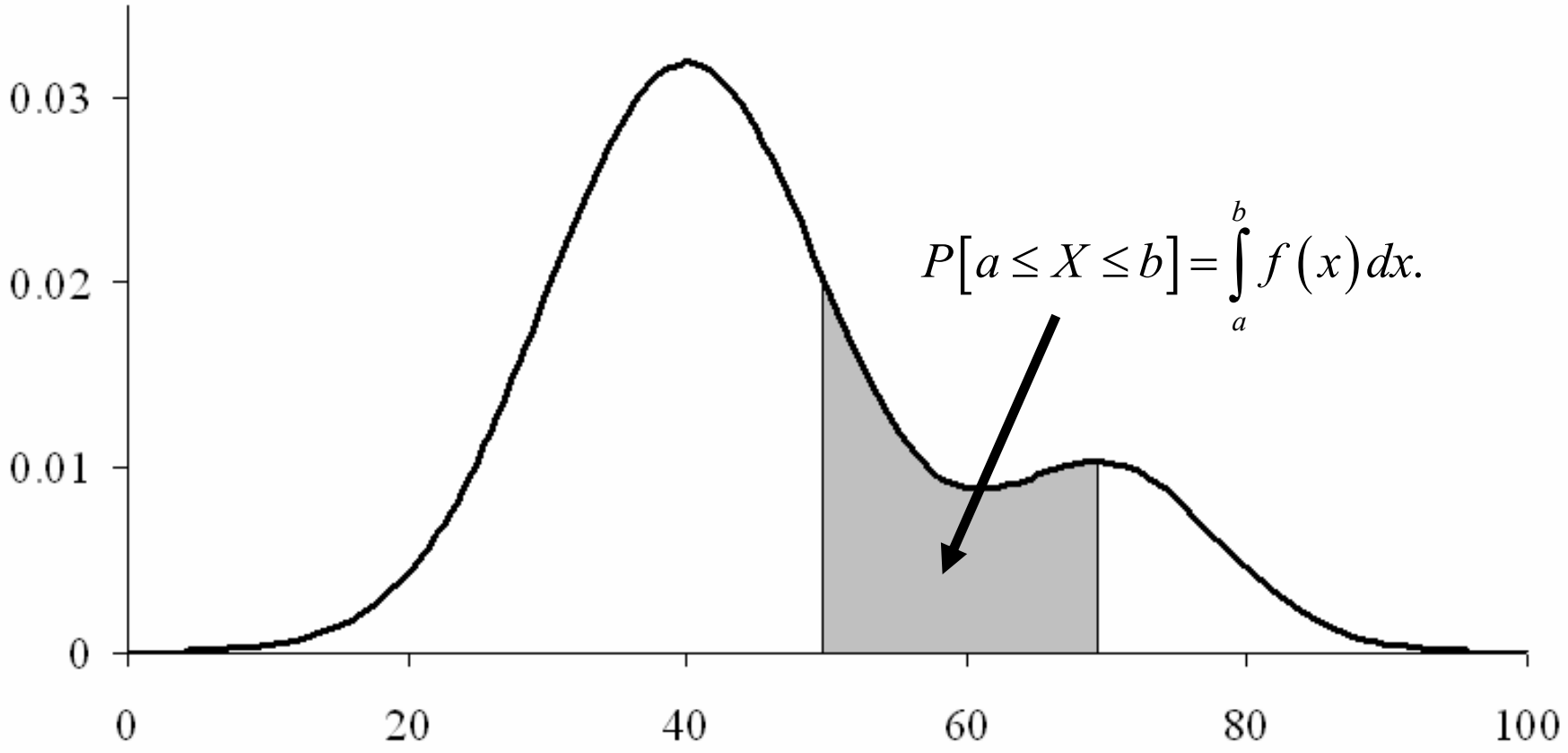
Let $f(x)$ denote a function defined for $-\infty < x < \infty$ with the following properties:

1. $f(x) \geq 0, \forall\, x \in \mathbb{R}$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. $P(a \leq X \leq b) = \int_{a}^{b} f(x)dx$

Then $f(x)$ is called the **probability density function (pdf)** of $X$. In this case the random, $X$, is called **continuous.**

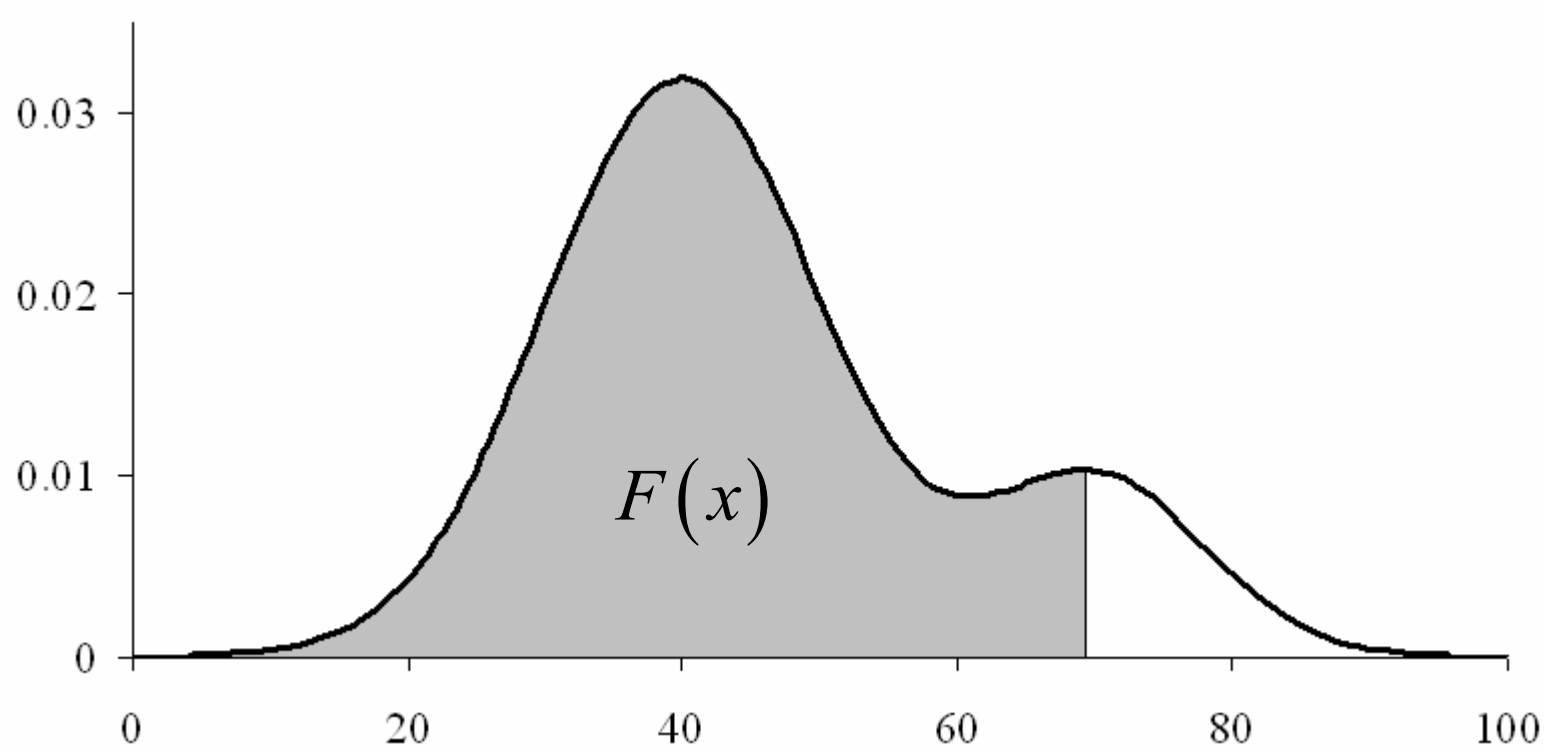
# Probability density function, $f(x)$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$



# Cumulative distribution function, $F(x)$

$$F(x) = P[X \le x] = \int_{-\infty}^{x} f(t)\,dt.$$



Thus, if $X$ is a continuous random variable with probability density function $f(x)$, then the cumulative distribution function of $X$ is given by:

$$F(x) = P[X \le x] = \int_{-\infty}^{x} f(t)\,dt.$$

Also because of the **fundamental theorem of calculus**.

$$F'(x) = \frac{dF(x)}{dx} = f(x)$$

# Continuous Random Variables

The cdf $F(x) = P(X \leq x)$ is a non-decreasing continuous function with

$$F(-\infty) = 0 \;\&\; F(\infty) = 1$$

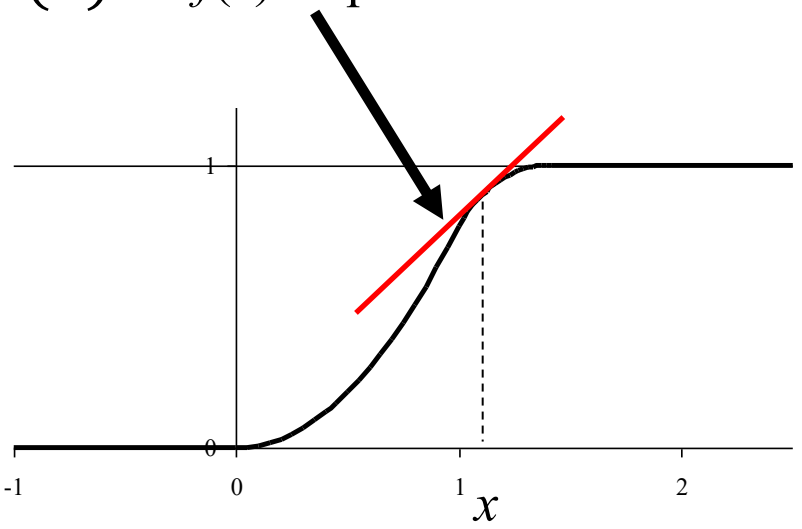
We say that a density (pdf) exists when

$$F(x) = \int_{-\infty}^{x} f(u)du$$

In these cases, $F(x)$ is differentiable and

$$f(x) = F'(x)$$

Note: There are continuous cdfs with "corners" at which they are not differentiable.



# Exercise

Suppose that

$$F_X(x) = \frac{1}{12}(x^2 + x^3)\,, x \in [0, 2]$$

Find $f(x)$.

# Example 2 (continued)

A point is selected at random from a square whose sides are of length 1. $X$ is the **distance** of the point from the lower left-hand corner.

$$F(x)=P[X\le x]=\begin{cases}0 & x\le 0\\ \dfrac{\pi x^2}{4} & 0\le x\le 1\\ \sqrt{x^2-1}+\left(\dfrac{\pi}{4}-\tan^{-1}\left(\sqrt{x^2-1}\right)\right)x^2 & 1\le x\le\sqrt{2}\\ 1 & \sqrt{2}\le x\end{cases}$$

Now we find the pdf by differentiation:

$$f(x)=F'(x)=\begin{cases}0, & x\le 0 \text{ or } x\ge\sqrt{2}\\ \dfrac{\pi x}{2}, & 0\le x\le 1\\ \dfrac{d}{dx}\left\{\sqrt{x^2-1}+\left[\dfrac{\pi}{4}-\tan^{-1}\left(\sqrt{x^2-1}\right)\right]x^2\right\}, & 1\le x\le\sqrt{2}\end{cases}$$

Now, to find the derivative:

$$\frac{d}{dx}\left\{\sqrt{x^2-1}+\left[\frac{\pi}{4}-\tan^{-1}\left(\sqrt{x^2-1}\right)\right]x^2\right\}$$

$$=\frac{1}{2}(x^2-1)^{-\frac{1}{2}}(2x)+\frac{\pi}{2}x\ -2x\tan^{-1}\left(\sqrt{x^2-1}\right)$$
$$-x^2\frac{d}{dx}\tan^{-1}\left(\sqrt{x^2-1}\right)$$

$$=\frac{\pi}{2}x+\frac{x}{(x^2-1)^{\frac{1}{2}}}-2x\tan^{-1}\left(\sqrt{x^2-1}\right)$$
$$-x^2\frac{d}{dx}\tan^{-1}\left(\sqrt{x^2-1}\right)$$

Now, recall that:

$$\frac{d}{du}\tan^{-1}u = \frac{1}{1+u^2}$$

$$\Rightarrow \frac{d}{dx}\tan^{-1}\left(\sqrt{x^2-1}\right) = \frac{1}{1+(x^2-1)}\left(\frac{1}{2}\right)(x^2-1)^{-½}(2x)$$

$$\Rightarrow -x^2\frac{d}{dx}\tan^{-1}\left(\sqrt{x^2-1}\right) = -\frac{x}{(x^2-1)^{\frac{1}{2}}}$$

$$\Rightarrow \frac{d}{dx}\left\{\sqrt{x^2-1}+\left[\frac{\pi}{4}-\tan^{-1}\left(\sqrt{x^2-1}\right)\right]x^2\right\}$$

$$= \frac{\pi}{2}x - 2x\tan^{-1}\left(\sqrt{x^2-1}\right)$$

Finally, the pdf is

$$f(x) = F'(x) = \begin{cases} 0, & if\ x \le 0\ or\ x \ge \sqrt{2} \\ \dfrac{\pi x}{2}, & if\quad x \in [0,1] \\ \dfrac{\pi x}{2} - 2x\arctan\left(\sqrt{x^2-1}\right), & if\ x \in [1,\sqrt{2}] \end{cases}$$

# Example 3

Suppose $X$ represents daily in-app purchase spending (in hundreds of dollars) by a mobile app user. A data science model captures spending behavior using a mixed random variable with:

- A **discrete point mass** at $x = 0$ (representing non-paying users): $P(X = 0) = 0.4$
- A **continuous density** for paying users over the range $0 < x \leq 2$: $f(x) = 0.3x$

a) Derive the Cumulative Distribution Function (CDF), $F(x)$.

b) Calculate the probability that a randomly chosen user spends at least \$100 ($X \geq 1$).

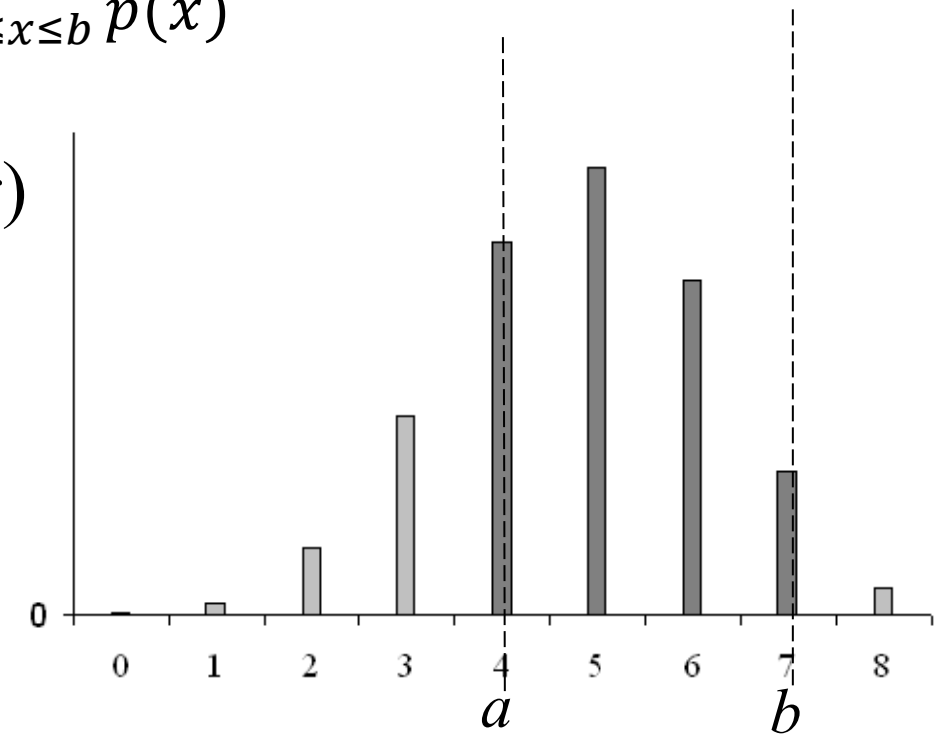
Answer for b): 0.45

# Discrete Random Variables

For a **discrete** random variable $X$ its distribution is described by the probability function, $p(x)$, which has the properties:

1. $0 \le p(x) \le 1$;
2. $\sum p(x) = 1$;
3. $P(a \le X \le b) = \sum_{a \le x \le b} p(x)$

Note: In the discrete case, $P(a \le X \le b)$ might be different than $P(a < X < b)$



# Example 4: Discrete Random Variable

Let $X$ have the following distribution:

| x | 3 | 5 | 8 | 10 |
|---|---|---|---|---|
| P(x) | 0.3 | 0.2 | 0.3 | ? |

Find

a) $P(X = 10) = 1 - (0.3 + 0.2 + 0.3) = 0.2$

b) $P(X \geq 5) = p(5) + p(8) + p(10) = 0.7$

c) $P(X > 5) = p(8) + p(10) = 0.5$

d) $P(4 < X \leq 8) = p(5) + p(8) = 0.5$

## The distribution function $F(x)$

This is defined for any random variable, $X$.

$$F(x) = P[X \leq x]$$

**Properties**

1. $F(-\infty) = 0$ and $F(\infty) = 1$.

Since $\{X \leq -\infty\} = \phi$ and $\{X \leq \infty\} = S$

then $F(-\infty) = 0$ and $F(\infty) = 1$.

2. $F(x)$ is a non-decreasing function. That is, if $x_1 < x_2$ then $F(x_1) \le F(x_2)$.

Proof:

If $x_1 < x_2$ then $\{X \le x_2\} = \{X \le x_1\} \cup \{x_1 < X \le x_2\}$

Thus $P[X \le x_2] = P[X \le x_1] + P[x_1 < X \le x_2]$

or $F(x_2) = F(x_1) + P[x_1 < X \le x_2]$

Since $P[x_1 < X \le x_2] \ge 0$ then $F(x_2) \ge F(x_1)$.

3. $F(b) - F(a) = P[a < X \le b]$.

If $a < b$ then using the argument above

$F(b) = F(a) + P[a < X \le b]$

Thus $F(b) - F(a) = P[a < X \le b]$.

4. $p(x) = P[X = x] = F(x) - F(x\text{-})$

Here $F(x-) = \lim_{u \to x-} F(u)$

5. If $p(x) = 0$ for all $x$ (i.e. $X$ is continuous) then $F(x)$ is continuous.

A function $F$ is continuous if

$$F(x-) = \lim_{u \to x-} F(u) = F(x+) = \lim_{u \to x+} F(u)$$

One can show that

Thus $p(x) = 0$ implies that $F(x+) = F(x) = F(x-)$

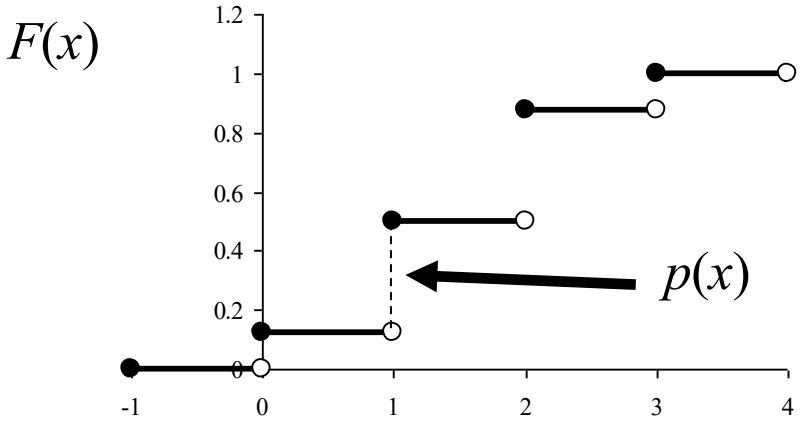
For Discrete Random Variables

$$F(x) = P[X \le x] = \sum_{u \le x} p(u)$$

$F(x)$ is a non-decreasing step function with

$$F(-\infty) = 0 \text{ and } F(\infty) = 1$$

$$p(x) = F(x) - F(x-) = \text{ jump in } F(x) \text{ at } x.$$



# Example 5: Discrete Random Variable

A data science team models the number of pages $X$ a user browses during a website session before exiting. The Probability Mass Function (PMF) is parameterized over an infinite discrete support ($x = 1,2,3, \ldots$) as:

$$p(x) = c \cdot x \cdot \left(\frac{1}{3}\right)^{x}, \quad \text{for } x \in \{1,2,3, \ldots\}$$

a) Find the normalization constant $c$ such that $p(x)$ is a valid PMF.

b) Calculate the probability that a user visits more than 2 pages, $P(X > 2)$.

## Example 3: Discrete Random Variable

Suppose we have two nontraditional dice. Die one has numbers 1, 2, 2, 3, 3, 4 and die two has numbers 1, 3, 4, 5, 6, 8. If both dice are rolled, what is the sample space? Let $X$ = sum of the two numbers showing. Derive the pmf of $X$.

# Example 3: Discrete Random Variable

Suppose we have two nontraditional dice. Die one has numbers 1, 2, 2, 3, 3, 4 and die two has numbers 1, 3, 4, 5, 6, 8. If both dice are rolled, what is the sample space? Let $X$ = sum of the two numbers showing. Derive the pmf of $X$.

| $x$ | 2 | 3 | .... | 12 |
|---|---|---|---|---|
| $p(x)$ | $\frac{1}{36}$ | $\frac{2}{36}$ | ..... | $\frac{1}{36}$ |

# Exercise : Discrete Random Variable

Five thousand lottery tickets are sold for \$1 each. One ticket will win \$1,000, two tickets will win \$500 each, and ten tickets will win \$100 each. Let $X$ denote the net gain from the purchase of a randomly selected ticket.

Construct the probability distribution of $X$